

No Peeking: Quantizing Medical Image Classifiers Without Calibration Data

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Abstract

Neural networks normally have their parameters stored in high-precision numerical form. Quantization is done to store the parameters with fewer bits. Many post-training quantization methods use a small set of input images to determine calibration statistics or rounding decisions. However, such images may not be available when a trained model is transferred. This study therefore examines how far weight-only compression can be applied when only the trained weights are available. We fine-tune MobileNetV3-Small on four MedMNIST datasets using three independent training seeds and evaluate weight-only compression without using images during the compression procedure. At 8 bits per weight, performance remains close to the FP32 baseline: mean AUC stays within 0.002 of FP32 across all four datasets, while the reported stored model size decreases from approximately 6.06 MB to 1.56 MB. The largest single-checkpoint AUC shift is 0.006. Below 8 bits, performance declines substantially. At 4 bits, we separate the effect of random rotation from the choice of scalar quantization grid using a matched 2×2 ablation. The experiment crosses rotation on or off with either a full-cardinality uniform grid or a Gaussian Lloyd–Max codebook. Testing all four combinations, rotation helps every time, but the highest mean AUC is obtained when rotation is combined with the full-cardinality uniform grid rather than the Gaussian codebook. Even that combination lands far behind the uncompressed model on accuracy. Which rotation is selected also matters: on PneumoniaMNIST, the mean within-checkpoint AUC standard deviation across five rotation seeds is 0.075. These numbers cover rounding error in the weights only. They say nothing about speed, memory use at inference, or clinical validity.

1 Introduction

Quantization lowers the precision used to store and run a network’s parameters [Gholami et al., 2022]. Most post-training quantization (PTQ) methods use calibration, in which a small set of input samples is passed through the model to estimate activation ranges or guide rounding decisions [Jacob et al., 2018; Nagel et al., 2020]. These calibration inputs may not be available when a trained model is transferred. Although calibration does not require labels, it still requires representative input images from the target distribution. This paper asks what is still possible without it.

Data-free weight-only quantization is evaluated on four MedMNIST v2 classification datasets [Yang et al., 2023]. The backbone is MobileNetV3-Small [Howard et al., 2019], chosen because it is already small, which leaves less slack for further compression. Each quantizer reads the trained weights and returns a low-rate stored form. At evaluation we rebuild FP32 weights from that stored form and then run inference, so the only thing that changes is weight rounding.

The low-bit method comes from incoherence processing. A random orthogonal rotation spreads the vector’s energy across its coordinates and can reduce the effect of large outliers before scalar quantization. QuIP uses rotations of this kind on LLM weights [Chee et al., 2023]. TurboQuant pairs a rotation with a Gaussian Lloyd–Max codebook for online vector quantization of embeddings and key-value caches [Zandieh et al., 2025]. We borrow that reconstruction mechanism and apply it to rows of convolutional weights. This is an adaptation, not a reimplement of TurboQuant’s deployment setting.

An earlier version of the experiment used different numbers of reconstruction levels for the two quantizers: the uniform grid used $2^b - 1$ levels, whereas the Gaussian codebook used 2^b centroids. Because both methods were charged the same bit rate, this comparison was not cardinality matched. The corrected experiment gives both quantization families the full 2^b reconstruction levels and evaluates them in a matched

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2×2 design that crosses rotation with scalar-grid choice. The layer policy and rotation setting are fixed before test evaluation, and no validation AUC is used to select the low-bit configuration. Rotation seeds are also kept independent of the training seeds, and none of these settings is selected using validation AUC. The study has three main findings:

1. Weight-only INT8 keeps mean AUC within 0.002 of FP32 across all four tasks, although the largest single-checkpoint AUC shift is 0.006.
2. At 4 bits, rotation improves the mean AUC of both scalar-grid families, but rotation combined with the full-cardinality uniform grid produces the highest mean AUC among the tested 4-bit configurations. At 2–3 bits, neither tested low-bit approach is reliable.
3. Rotation seed contributes substantial variation even when the training seed is held fixed, showing that a single rotation can influence the ranking of low-bit methods.

2 Related Work

PTQ methods differ mostly in what they are allowed to look at. AdaRound and BRECQ learn rounding from calibration activations [Nagel et al., 2020; Li et al., 2021]. GPTQ uses calibration data for second-order error correction [Frantar et al., 2023], and AWQ and SmoothQuant use measured activation statistics [Lin et al., 2024; Xiao et al., 2023]. These methods operate under less restrictive data-access assumptions than those used in this study.

Data-free quantization is not only direct rounding.

Weight equalization and bias correction improve 8bit vision models with no data at all [Nagel et al., 2019]. Another line synthesizes calibration inputs from the model’s own batch-norm statistics [Cai et al., 2020]. We exclude synthetic-data methods because the present protocol permits only transformations of the stored weights. OptRot is the closest recent work, learning weight rotations without calibration data [Gadhikar et al., 2025], though its rotations are designed to fold into LLM architectures. Our scope is narrower: fixed random rotations used as a storage transform on compact CNN weights.

MedMNIST supplies standardized, small biomedical tasks with fixed train, validation, and test splits [Yang et al., 2023]. That makes compression experiments easy to reproduce. It does not come close to the resolution, population shift, or validation burden of a real clinical system.

Table 1: MedMNIST v2 datasets used in the study. “Train/Val/Test” reports the number of images in each split. Imbalance is the ratio of the largest to smallest training-class size.

Dataset	Classes	Train/Val/Test	Imbalance
DermaMNIST	7	7007/1003/2005	58.7:1
PneumoniaMNIST	2	4708/524/624	2.9:1
BloodMNIST	8	11959/1712/3421	2.7:1
PathMNIST	9	89996/10004/7180	1.6:1

3 Methods

3.1 Datasets and Checkpoints

Table 1 gives the four MedMNIST v2 datasets used in this experiment: DermaMNIST, PneumoniaMNIST, BloodMNIST, and PathMNIST. Images are uniformly resized to 64×64 pixels, and normalization is done according to the statistics of ImageNet channels. For grayscale images, the average of the three ImageNet channel means is considered. Since DermaMNIST is severely imbalanced, AUC is considered as the main evaluation metric, with accuracy and macro F_1 as additional metrics. MobileNetV3-Small is pretrained on ImageNet and fine-tuned independently for each dataset, using the three seed values for models 42, 123, and 456. In the case of grayscale inputs, the weights for the first RGB convolution layer are averaged along the input-channel dimension and replaced by the classifier head for different numbers of classes in each task. Training uses AdamW with cosine learning-rate decay, with validation AUC monitored during training. After training, the resulting 12 checkpoints are fixed and used for all compression experiments; no subsequent compression configuration is selected using validation performance.

3.2 Weight Rows and Stored Rate

A convolutional tensor of shape (o, c, k_h, k_w) is reshaped to $W \in \mathbb{R}^{o \times n}$, where $n = ck_hk_w$. Linear weights already look like this. Each output row $\mathbf{w} \in \mathbb{R}^n$ is quantized on its own. We include only convolution and linear tensors with at least 256 parameters. Biases, batch-normalization parameters, and excluded tensors are stored in FP16.

The first convolution and final classifier remain at INT8 for all 2-, 3-, and 4-bit experiments. This layer policy is fixed before test evaluation and is applied consistently across datasets. Convolutional and linear tensors with fewer than 256 parameters are excluded from low-bit quantization, while biases, batch-normalization parameters, and other excluded tensors are stored in FP16. For eligible weight rows with

Table 2: Matched 2×2 ablation. Every cell uses 2^b reconstruction levels and one FP16 scale per row.

	Uniform grid	Gaussian codebook
No rotation	Uniform	Gaussian
Rotation	Rot.+uniform	Rot.+Gaussian

fan-in $n < 64$, rotation is disabled while the selected scalar quantization grid remains unchanged. Thus, rotation is determined by the predefined layer and fan-in policy and is not selected using validation performance. The four 4-bit ablation cells use the same quantization-rate definition and have a mean stored size of approximately 0.810 MB, compared with 6.06 MB for the FP32 checkpoints.

3.3 Matched Scalar Grids

For every bit width b , both scalar-quantization families use exactly $L = 2^b$ reconstruction levels. This matched cardinality ensures that differences between the uniform and Gaussian families are not caused by different numbers of reconstruction levels.

Uniform grid. For $L = 2^b$, let $a = \max_i |w_i|$, $s = a/(L - 1)$, and

$$\mathcal{C}_b = \{-(L - 1), -(L - 3), \dots, L - 3, L - 1\}.$$

Each weight is reconstructed as

$$\hat{w}_i = s \arg \min_{c \in \mathcal{C}_b} |w_i/s - c|.$$

This mid-rise grid is symmetric and uses all L bit patterns. A symmetric grid with an even number of levels has no exact zero, so exact zeros fall to a fixed tie rule. One FP16 scale is stored per row.

Gaussian codebook. Let $s = \|\mathbf{w}\|_2/\sqrt{n}$. Each weight row is normalized by this scale, and each coordinate is assigned to the nearest of $L = 2^b$ Lloyd–Max centroids fitted once to a standard normal distribution [Lloyd, 1982; Max, 1960]. The same codebook is shared across rows and datasets at a given bit width, so no dataset-specific calibration is required.

3.4 Rotation and the 2×2 Ablation

For a row vector, draw a Haar-random orthogonal matrix $\Pi \in \mathbb{R}^{n \times n}$ and compute

$$\mathbf{y} = (\mathbf{w}/s)\Pi^\top.$$

A random orthogonal rotation preserves the vector norm and makes the coordinate distribution approach

a standard normal distribution as n increases. After scalar quantization we reconstruct with

$$\hat{\mathbf{w}} = s \hat{\mathbf{y}} \Pi.$$

The rotation is regenerated deterministically from a dedicated rotation seed. In the main 2-, 3-, and 4-bit experiments, rotation seed 0 is fixed across checkpoints, while a deterministic tensor index generates a distinct rotation for each eligible weight tensor. The training seed is never reused to construct a rotation. This separation prevents training-seed variation from being confounded with variation caused by the rotation transform.

Accordingly, the main low-bit evaluation uses a pre-determined rotation setting and rotation seed rather than selecting either quantity after observing validation or test performance.

Table 2 defines the resulting matched 4-bit 2×2 ablation. The two factors, rotation status and scalar-grid family, are crossed independently, and all four cells use the same reconstruction-level count, layer policy, fan-in threshold, and rate definition.

3.5 Evaluation Protocol

The compression evaluation is strictly data-free: no validation or test images are used to select quantization settings, layer policies, rotation counts, or rotation seeds. For the low-bit experiments, the evaluated bit widths are $b \in \{2, 3, 4\}$. The layer policy, fan-in threshold, scalar-grid family, rotation setting, and main rotation seed are fixed before test evaluation and applied consistently across all four datasets. Test AUC, accuracy, and macro- F_1 are reported as the mean and sample standard deviation across the three model-training seeds, with method differences paired at the checkpoint level.

INT8 is evaluated separately as a per-channel symmetric weight-only baseline. The INT8 comparison is not used to select any configuration for the low-bit experiments; it provides a higher-precision reference for assessing the effect of weight rounding and storage reduction. The same fixed checkpoints are used for the FP32 and INT8 comparison.

A separate factorial experiment evaluates the 4-bit rotated Gaussian configuration using rotation seeds $\{0, 1, 2, 3, 4\}$ on each of the same 12 fixed checkpoints. Holding the trained checkpoint constant while changing only the rotation seed separates training-related variation from rotation-related variation. The resulting analysis is descriptive rather than a formal hypothesis test because only three independent training seeds are available.

Table 3: Test AUC, accuracy, and stored model size for FP32 and per-channel symmetric INT8 across the four datasets. Metrics are reported over the three model seeds; size is reported in MB.

Dataset	Precision	AUC	Accuracy	Size
DermaMNIST	FP32	0.917	0.765	6.07
	INT8	0.917	0.762	1.56
PneumoniaMNIST	FP32	0.958	0.847	6.05
	INT8	0.959	0.845	1.56
BloodMNIST	FP32	0.998	0.971	6.07
	INT8	0.998	0.970	1.56
PathMNIST	FP32	0.991	0.898	6.08
	INT8	0.990	0.895	1.56

Quantization is implemented as a fake-quantization evaluation: stored quantization indices and associated scales are reconstructed into FP32 weights before inference. Thus, the results measure the effect of weight rounding rather than integer-kernel execution, inference latency, or runtime memory consumption. Reported model sizes account for weight indices, row scales, and higher-precision parameters included by the storage representation.

No calibration images are used at any stage of the compression evaluation. The compression procedure operates only on the fixed trained weights and predetermined quantization settings; validation AUC is not used to choose the layer policy, bit width, number of rotations, or rotation seed.

4 Results

4.1 INT8 Keeps the Checkpoint Metrics

Table 3 compares the FP32 checkpoints with the per-channel symmetric INT8 baseline across the four datasets. INT8 preserves the ranking performance of the FP32 models closely: mean AUC differs by no more than 0.002 on any dataset, with the largest single-checkpoint shift of 0.006 occurring on PneumoniaMNIST. Mean stored model size decreases from approximately 6.06 MB for FP32 to 1.56 MB for INT8, corresponding to a 3.9-fold reduction. FP16 also reproduces the FP32 metrics closely and requires approximately 3.05 MB.

The FP32 and INT8 comparisons use the same 12 fixed checkpoints, with three independently trained checkpoints for each dataset. Thus, the reported dataset-level means reflect the three training seeds and do not involve additional model selection after compression. The reported dataset-level means therefore summarize the three model seeds rather than

introducing additional model selection after compression. The largest observed single-checkpoint AUC difference is 0.006, while the dataset-level mean differences remain within 0.002.

4.2 Synthetic Reconstruction Separates Rotation from Grid Choice

Figure 1 evaluates reconstruction error before the downstream experiments. On Gaussian weights, rotation changes the reconstruction error only modestly because the source distribution already resembles the Gaussian codebook. On heavy-tailed weights, rotation reduces normalized reconstruction error for both quantization families. In this controlled synthetic experiment, the Gaussian codebook produces lower reconstruction error than the uniform grid across the tested settings.

4.3 At 4 Bits, Rotation Helps More Reliably Than the Gaussian Grid

Table 4 and Figure 2 report the matched 4-bit 2×2 comparison. Rotation plus uniform achieves the highest mean AUC among the four configurations on all four datasets. Relative to unrotated uniform quantization, the mean AUC increases by 0.059 on DermaMNIST, 0.155 on PneumoniaMNIST, 0.041 on BloodMNIST, and 0.069 on PathMNIST. The paired improvement is positive on 8 of the 12 checkpoints, indicating that the effect is not uniform across individual trained models.

Rotation also improves the Gaussian codebook on all 12 paired checkpoints. Nevertheless, the rotated Gaussian configuration does not achieve the highest mean AUC on any of the four datasets; rotation plus uniform has the highest mean AUC in every dataset-level comparison. Thus, within this experiment, the results support a benefit from rotation at 4 bits, but they do not support the claim that the Gaussian Lloyd-Max codebook provides a downstream advantage over the matched uniform grid.

Higher AUC does not necessarily imply that the quantized model remains a reliable classifier. For rotation plus uniform, mean accuracy ranges from 0.50 on BloodMNIST to 0.76 on PneumoniaMNIST, while mean macro- F_1 ranges from 0.20 on DermaMNIST to 0.68 on PneumoniaMNIST. The corresponding FP32 mean accuracy values range from 0.77 to 0.97. Thus, some low-bit models retain part of their ranking ability while showing substantially weaker class-level predictions, particularly on the multiclass tasks..

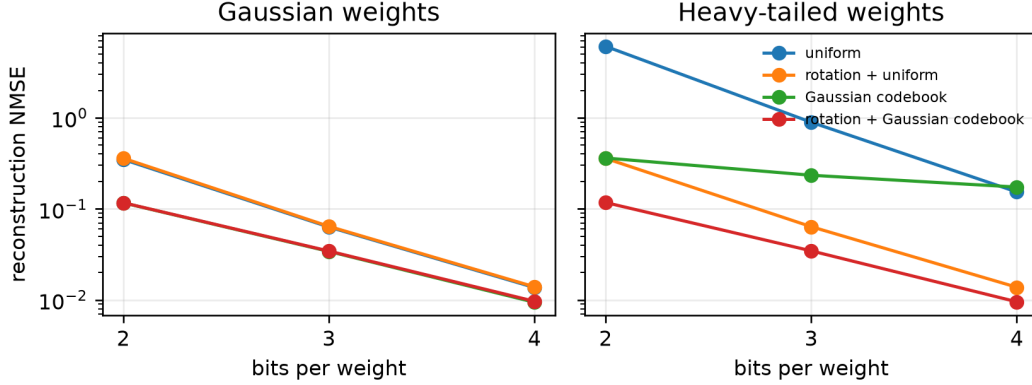


Figure 1: Normalized reconstruction MSE for uniform and Gaussian scalar quantization on independently generated Gaussian and Student- t weight matrices at 2, 3, and 4 bits. The rotated configurations use an independently assigned rotation seed.

Table 4: Four-bit test AUC, reported as mean $\pm$ sample standard deviation over three model seeds. The table is the matched 2×2 ablation crossing rotation status with scalar-grid family; all cells use the same reconstruction-level count, layer policy, fan-in threshold, and rate definition, with rotation seed fixed where rotation is enabled.

Dataset	Uniform	Rotation + uniform	Gaussian	Rotation + Gaussian
DermaMNIST	0.614 $\pm$ 0.024	0.673$\pm$0.040	0.546 $\pm$ 0.068	0.595 $\pm$ 0.070
PneumoniaMNIST	0.685 $\pm$ 0.085	0.840$\pm$0.054	0.689 $\pm$ 0.129	0.791 $\pm$ 0.139
BloodMNIST	0.771 $\pm$ 0.018	0.812$\pm$0.092	0.631 $\pm$ 0.097	0.796 $\pm$ 0.029
PathMNIST	0.821 $\pm$ 0.098	0.890$\pm$0.053	0.741 $\pm$ 0.055	0.754 $\pm$ 0.065

4.4 Two and Three Bits Are Not Reliable

Figure 3 summarizes the fixed uniform and rotation-plus-Gaussian configurations across 2-, 3-, 4-, and 8-bit settings. When pooled over the 12 checkpoints, 2-bit AUC is 0.498 for uniform quantization and 0.466 for rotation plus Gaussian. At 3 bits, the corresponding pooled AUC values are 0.505 and 0.521. At 4 bits, pooled AUC improves to 0.723 for uniform and 0.734 for rotation plus Gaussian, but these values remain substantially below the INT8 pooled AUC of 0.966.

These pooled results indicate that 2- and 3-bit quantization are not reliable under the fixed protocol. Although the 4-bit configurations recover substantially more ranking ability, their pooled AUC remains well below the INT8 reference. The results therefore support INT8 as the robust low-storage operating point in this experiment, while 4-bit quantization remains sensitive to the chosen configuration and model checkpoint.

4.5 Rotation Seed Is Its Own Source of Variation

Figure 4 holds each trained checkpoint fixed while changing only the rotation seed. PneumoniaMNIST shows the largest within-checkpoint variation, with a mean standard deviation of 0.075 AUC across its three checkpoints. For the checkpoint trained with seed 42, AUC ranges from 0.608 to 0.930 across the five rotation seeds. The corresponding mean within-checkpoint standard deviations are 0.025 for DermaMNIST, 0.029 for BloodMNIST, and 0.022 for PathMNIST.

Training-seed variation is also substantial. Averaged over rotation seeds, the between-model standard deviation is 0.078 on DermaMNIST, 0.076 on PneumoniaMNIST, 0.041 on BloodMNIST, and 0.063 on PathMNIST. These results indicate that both training and rotation seeds contribute meaningful variation near the low-bit performance range.

The 4-bit ablation separates reconstruction behavior from downstream classification behavior. In the synthetic experiment, the Gaussian codebook produces lower reconstruction error than the uniform grid. In the four-dataset 4-bit evaluation, however, rotation plus uniform achieves the highest mean AUC in every dataset-level comparison. This difference in-

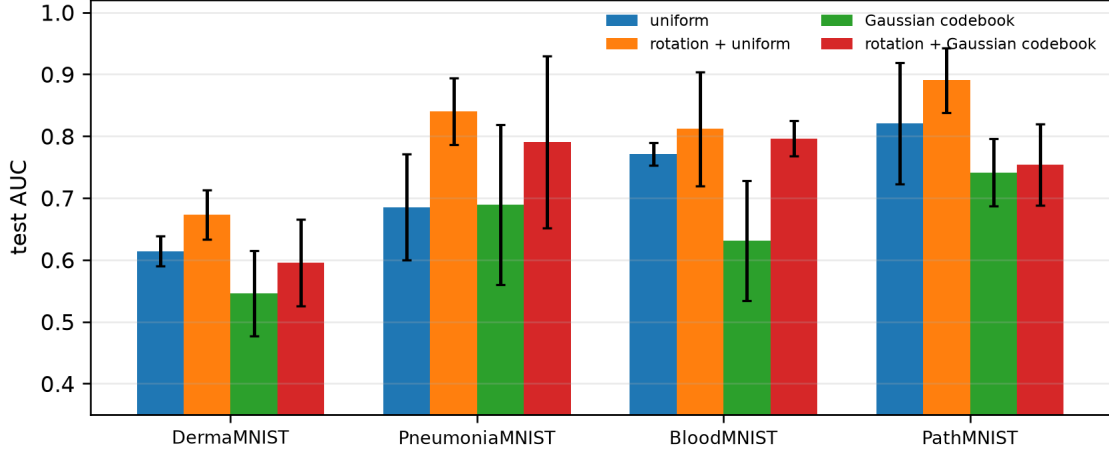


Figure 2: Matched 4-bit 2×2 ablation crossing rotation status with scalar-grid family. Bars show mean test AUC over three model seeds, and error bars show the corresponding sample standard deviation. Rotation + uniform has the highest mean AUC on all four datasets.

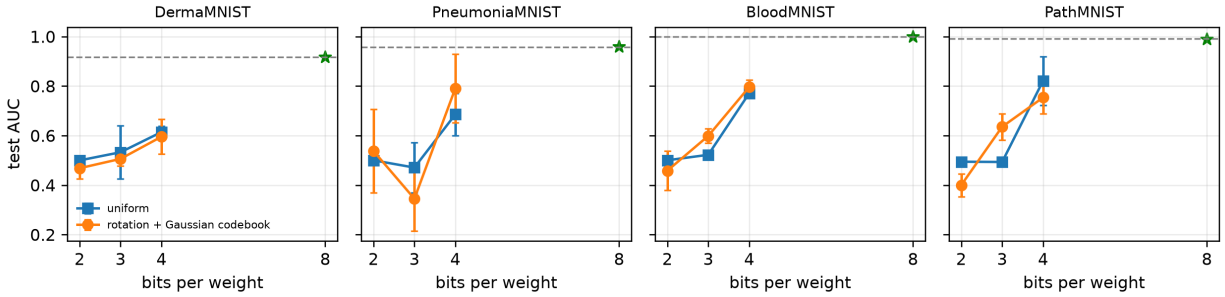


Figure 3: Test AUC under the fixed protocol. Dashed lines indicate the FP32 reference and stars indicate the INT8 reference. Curves show mean $\pm$ sample standard deviation over the three model seeds.

indicates that reconstruction error alone is not sufficient to predict downstream classification behavior in the tested setting. Weight-space MSE therefore does not fully predict downstream classification performance. The result also shows why mean $\pm$ sample standard deviation should be reported across model seeds.

The matched baselines are important because crossing rotation status with scalar-grid family allows the downstream contribution of each factor to be examined separately.

The factorial experiment says something direct about rotation sensitivity. The rotation-seed experiment also shows that the choice of rotation can materially affect downstream AUC. Selecting a rotation using a data-free reconstruction criterion might reduce this variation, but such a selection rule was not evaluated in the present study. Any proposed data-free selection criterion would therefore require separate validation against paired downstream performance.

4.6 Limitations

The study uses one backbone, one image resolution, four benchmark datasets, three training seeds, and five rotation seeds. MedMNIST images are downsampled and standardized, so nothing here validates a medical device or a clinical decision rule. The data-free setting is most relevant when representative calibration inputs are unavailable. Institutions that can calibrate locally may not face this constraint.

Only weights are quantized. Activations remain in floating point, and the quantized weights are reconstructed to FP32 before inference. Therefore, the reported stored size is a representation estimate, not a measurement of runtime memory. Dense Haar rotations also cost more decode work than structured Hadamard transforms. Finally, test-set bootstrap intervals were not calculated. The reported uncertainty therefore reflects model and rotation seeds but not variation from finite test-set sampling.

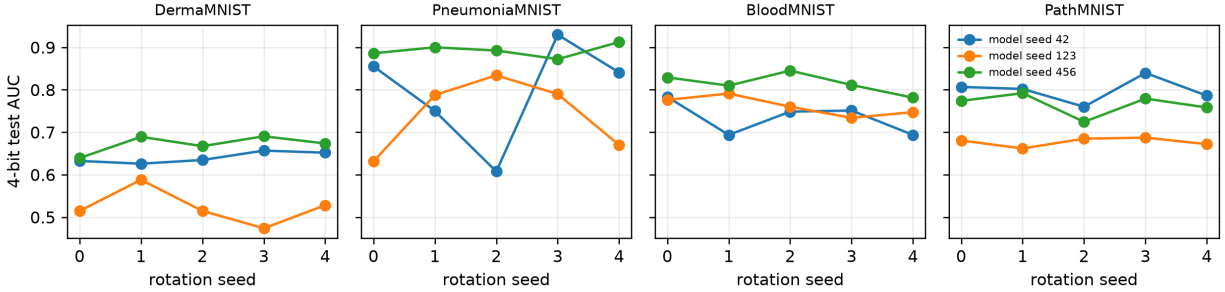


Figure 4: Four-bit rotation plus Gaussian codebook evaluated over three model-training seeds and five independently assigned rotation seeds. Each line holds one trained checkpoint fixed while only the rotation seed changes.

5 Conclusion

Under the fixed, calibration-free protocol, INT8 weight rounding preserves mean AUC closely relative to FP32 across the four MedMNIST datasets. At 2 and 3 bits, the tested configurations show unreliable performance, while 4-bit quantization recovers part of the ranking performance but remains substantially below INT8. In the matched 4-bit 2×2 experiment, rotation plus uniform quantization achieves the highest mean AUC on all four datasets.

The rotation-seed and training-seed experiments also show that both sources of variation can materially affect low-bit results. These findings support matched comparisons and multi-seed evaluation when studying low-bit quantization. They describe weight-rounding behavior under the specified experimental protocol and do not establish clinical validity, deployment performance, runtime-memory savings, or latency improvements.

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